GPS Spoofing as a Defensive Countermeasure Against an Autonomous Enemy Drone Attack:

Closed-Loop Trajectory Redirection, Inertial Residual Analysis, and Minimum-Harm Safe-Zone Optimization

Authors

Parth Chavan · Maincy Kumawat Naman Girdhani
Pranav Borole · Pritam Dahiya · Rahul Kumawat

Under the supervision of ***Dr. Varun Kumar***
Department of Electronics and Communication Engineering, IIIT Vadodara

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Abstract

Unmanned Aerial Vehicles (UAVs) relying on unencrypted civilian Global Positioning System (GPS) signals for waypoint navigation are vulnerable to structured signal-deception countermeasures. Unlike kinetic interception, which risks falling debris over densely populated urban areas, *defensive GPS spoofing* injects a controlled, time-varying position bias $\boldsymbol{\delta}[k]$ into the drone's receiver so that the vehicle's own closed-loop guidance law steers its physical trajectory into a pre-selected low-harm safe zone $\mathbf{x}_{\text{safe}}$ while its onboard navigation computer believes it is flying straight toward the protected city $\mathbf{x}_{\text{city}}$. However, modern autonomous drones cross-check GPS observations against an internal Inertial Measurement Unit (IMU) dead-reckoning estimator and trigger an anomaly alarm whenever the innovation residual $r[k]$ exceeds a threshold τ. In this paper, we formulate a complete discrete-time mathematical model of this cyber-physical tug-of-war, prove the exact steady-state condition $\boldsymbol{\delta}_{\text{final}} = \mathbf{x}_{\text{city}} - \mathbf{x}_{\text{safe}}$ via a virtual ghost-target transformation, derive the closed-form relationship $r_{\text{ss}} \approx \Delta\delta_{\max}/\alpha_{\text{INS}}$ governing the stealth ramp rate, and couple the controller to a 2D Gaussian population harm cost map $H(x, y)$. Simulation experiments across baseline, stealth-spoofed, aggressive-jump, and multi-drone scenarios demonstrate that rate-limiting the spoofing offset to $\Delta\delta^*_{\max} = 5.57$ m/step (11.14 m/s) diverts a 20 m/s attacking drone from a 100/100 harm city core into a 3.42/100 minimum-damage basin with a terminal miss distance of 0.40 m while bounding the peak detector residual at 84.12 m $< \tau = 90.0$ m.

Keywords: Defensive GPS Spoofing, Autonomous UAV Redirection, Inertial Dead-Reckoning Residual, Stealth Rate-Limit Optimization, Spatial Harm Cost Map, Closed-Loop Guidance Deception.

1 Introduction: Problem Formulation and Project Scope (What We Did)

Autonomous drones navigating toward fixed geographic coordinates pose a severe security challenge when targeting densely populated metropolitan areas. Kinetic countermeasures (such as surface-to-air interceptors or net guns) are often impractical over urban centers because kinetic impact or loss of rotor thrust causes the airframe and payload to fall directly onto high-density civilian infrastructure. Consequently, *non-kinetic electronic redirection* via defensive GPS spoofing offers a compelling alternative: rather than destroying the drone mid-air, the defender overrides the ambient satellite navigation signal with a synthesized GPS broadcast that smoothly bends the drone's flight path into an uninhabited safe zone (such as an open lake or rural field).

To investigate this problem rigorously, we designed and implemented a complete, reproducible discrete-time Python simulation framework (`simulation.py`) that models the interaction between three distinct agents:

1. **The Enemy Autonomous Drone (`Drone` class)**: A 2D point-mass vehicle launched from an arbitrary user-specified coordinate $\mathbf{x}_{\text{start}} = [x_0, y_0]^T$ that steers at maximum cruise speed $v_{\max} = 20.0$ m/s toward a user-specified protected city center $\mathbf{x}_{\text{city}} = [x_c, y_c]^T$ using proportional line-of-sight guidance driven solely by its perceived GPS coordinate $\mathbf{x}_{\text{perceived}}[k]$.
2. **The Onboard Anomaly Detector (`Detector` class)**: An internal fault-detection monitor that propagates an aided strapdown Inertial Measurement Unit (IMU) dead-reckoning state $\mathbf{x}_{\text{imu}}[k]$ subject to random-walk sensor drift $\mathbf{w}_{\text{imu}}[k]$ and computes the Euclidean innovation residual $r[k] = \|\mathbf{x}_{\text{perceived}}[k] - \mathbf{x}_{\text{imu}}[k]\|_2$. If $r[k]$ exceeds the detection threshold $\tau = 90.0$ m, the drone flags a spoofing attack.
3. **The Defensive GPS Spoofer & Spatial Risk Optimizer (`Spoofer` class)**: A defensive system that constructs a 2D population/infrastructure Harm Cost Map $H(x, y) \in [1, 100]$ centered on the user's city coordinate (representing the darkest, maximum-harm zone on the map), automatically locates the reachable coordinate of *minimum collateral damage* $\mathbf{x}_{\text{safe}} = \arg\min H(x, y)$, and injects a rate-limited spoofing ramp $\boldsymbol{\delta}[k]$ that diverts the drone to $\mathbf{x}_{\text{safe}}$ without ever violating $r[k] \leq \tau$.

Core Contribution of Our Work: We bridge spatial risk minimization with closed-loop sensor deception: given *any* user-entered drone launch site $\mathbf{x}_{\text{start}}$ and city center $\mathbf{x}_{\text{city}}$, our algorithm automatically synthesizes the surrounding harm topography, selects the minimum-damage landing zone $\mathbf{x}_{\text{safe}}$, solves for the fastest undetectable spoofing ramp rate $\Delta\delta^*_{\max}$, and diverts both single-drone and multi-drone formations with sub-meter terminal accuracy.

2 System Architecture and Discrete-Time Mathematical Model (How We Did It)

The simulation operates on a uniform discrete time grid $t_k = k\Delta t$ for $k \in \{0, 1, \ldots, N\}$, where $\Delta t = 0.5$ s is the sampling interval. At each discrete time step k, the closed-loop system executes seven coupled mathematical equations.

2.1 2D Point-Mass Drone Kinematics (Equation 1)

Let $\mathbf{x}_{\text{true}}[k] = [x_{\text{true}}[k],\ y_{\text{true}}[k]]^T \in \mathbb{R}^2$ denote the drone's true physical position in a 2D East-North Cartesian frame ($[0, 1000]$ m $\times$ $[0, 1000]$ m), and let $\mathbf{v}[k] = [v_x[k],\ v_y[k]]^T \in \mathbb{R}^2$ denote its velocity vector. Using first-order forward Euler integration, the physical state evolves as:

$$\mathbf{x}_{\text{true}}[k+1] = \mathbf{x}_{\text{true}}[k] + \mathbf{v}[k]\,\Delta t, \qquad \mathbf{x}_{\text{true}}[0] = \mathbf{x}_{\text{start}}. \tag{1}$$

At maximum cruise speed $v_{\max} = 20.0$ m/s and step size $\Delta t = 0.5$ s, the drone advances by $v_{\max}\Delta t = 10.0$ m per discrete step.

2.2 Spoofed GPS Observation Model (Equation 2)

The drone's navigation receiver does not observe $\mathbf{x}_{\text{true}}[k]$ directly. Instead, it measures a corrupted GPS position vector $\mathbf{x}_{\text{perceived}}[k]$ (denoted $\mathbf{z}_k$ in estimation literature) given by:

$$\mathbf{x}_{\text{perceived}}[k] = \mathbf{x}_{\text{true}}[k] + \boldsymbol{\delta}[k] + \boldsymbol{\eta}[k], \qquad \boldsymbol{\eta}[k] \sim \mathcal{N}(\mathbf{0},\ \sigma^2_{\text{gps}}\mathbf{I}_2), \tag{2}$$

where $\boldsymbol{\delta}[k] = [\delta_x[k],\ \delta_y[k]]^T \in \mathbb{R}^2$ is the deterministic spoofing offset vector injected by the defender at step k, and $\boldsymbol{\eta}[k]$ is zero-mean bivariate Gaussian receiver noise with standard deviation $\sigma_{\text{gps}} = 1.2$ m per axis.

2.3 Enemy Proportional Line-of-Sight Guidance Law (Equation 3)

The drone's flight controller is programmed to fly directly toward the protected city $\mathbf{x}_{\text{city}}$. Because it trusts its GPS receiver, it computes the perceived line-of-sight error vector $\mathbf{e}_{\text{perc}}[k] = \mathbf{x}_{\text{city}} - \mathbf{x}_{\text{perceived}}[k]$ and aligns its velocity vector along the unit direction pointing toward $\mathbf{x}_{\text{city}}$:

$$\mathbf{v}[k] = s\big(\|\mathbf{e}_{\text{perc}}[k]\|_2\big)\,\frac{\mathbf{x}_{\text{city}} - \mathbf{x}_{\text{perceived}}[k]}{\|\mathbf{x}_{\text{city}} - \mathbf{x}_{\text{perceived}}[k]\|_2}, \qquad s(d) = v_{\max}\min\left(1,\ \frac{d}{R_{\text{prop}}}\right). \tag{3}$$

Here, $s(d) = v_{\max} = 20.0$ m/s throughout the cruise phase ($d \geq R_{\text{prop}} = 35.0$ m), matching the unit-vector guidance law $\mathbf{v}[k] = v_{\max}(\mathbf{x}_{\text{city}} - \mathbf{x}_{\text{perceived}}[k])/\|\mathbf{x}_{\text{city}} - \mathbf{x}_{\text{perceived}}[k]\|_2$. Inside the small terminal capture radius $R_{\text{prop}} = 35.0$ m, the speed tapers proportionally with distance d so that the discrete-time vehicle settles smoothly onto its target without limit-cycle overshoot (10 m step hopping).

2.4 Rate-Limited Stealth Spoofing Offset Design (Equation 4)

To divert the drone from $\mathbf{x}_{\text{city}}$ to the chosen safe zone $\mathbf{x}_{\text{safe}}$, the spoofer initializes $\boldsymbol{\delta}[0] = \mathbf{0}$ and drives $\boldsymbol{\delta}[k]$ toward the terminal offset vector:

$$\boldsymbol{\delta}_{\text{final}} = \mathbf{x}_{\text{city}} - \mathbf{x}_{\text{safe}}. \tag{4}$$

To prevent an instantaneous position jump from alerting the onboard detector, the per-step change $\Delta\boldsymbol{\delta}[k] = \boldsymbol{\delta}[k+1] - \boldsymbol{\delta}[k]$ is constrained by a maximum stealth step rate $\Delta\delta_{\max}$ (in m/step, corresponding to $v_\delta = \Delta\delta_{\max}/\Delta t$ in m/s):

$$\boldsymbol{\delta}[k+1] = \boldsymbol{\delta}[k] + \min\Big(\Delta\delta_{\max},\ \|\boldsymbol{\delta}_{\text{final}} - \boldsymbol{\delta}[k]\|_2\Big)\,\frac{\boldsymbol{\delta}_{\text{final}} - \boldsymbol{\delta}[k]}{\|\boldsymbol{\delta}_{\text{final}} - \boldsymbol{\delta}[k]\|_2}, \qquad \|\boldsymbol{\delta}[k+1] - \boldsymbol{\delta}[k]\|_2 \leq \Delta\delta_{\max}. \tag{5}$$

2.5 Aided Inertial Estimator and Residual Threshold Detector (Equation 5)

An open-loop strapdown IMU that integrates accelerometer readings accumulates unbounded position error due to sensor bias and random-walk noise. In realistic UAV autopilots, the dead-reckoning state $\mathbf{x}_{\text{imu}}[k]$ is propagated using the commanded velocity $\mathbf{v}[k]$, Gaussian random-walk drift $\mathbf{w}_{\text{imu}}[k] \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{imu}}^2\mathbf{I}_2)$ ($\sigma_{\text{imu}} = 0.25$ m), and a weak complementary drift-bounding feedback gain $\alpha_{\text{INS}} = 0.068$:

$$\mathbf{x}_{\text{imu}}[k+1] = \mathbf{x}_{\text{imu}}[k] + \mathbf{v}[k]\,\Delta t + \alpha_{\text{INS}}\big(\mathbf{x}_{\text{perceived}}[k] - \mathbf{x}_{\text{imu}}[k]\big) + \mathbf{w}_{\text{imu}}[k]. \tag{6}$$

At every step k, the onboard anomaly detector computes the Euclidean innovation residual $r[k]$ and evaluates a binary threshold decision rule $D[k]$ against $\tau = 90.0$ m:

$$r[k] = \big\|\mathbf{x}_{\text{perceived}}[k] - \mathbf{x}_{\text{imu}}[k]\big\|_2, \qquad D[k] = \begin{cases} 1 & \text{(Spoofing Alarm Triggered)}, \quad \text{if } r[k] > \tau, \\ 0 & \text{(Nominal / Stealth Maintained)}, \quad \text{if } r[k] \leq \tau. \end{cases} \tag{7}$$

2.6 2D Gaussian Harm Cost Map and Safe-Zone Optimization (Equation 6)

Given the user-specified drone launch point $\mathbf{x}_{\text{start}}$ and protected city coordinate $\mathbf{x}_{\text{city}}$, we define a continuous 2D spatial harm index $H(x, y) \in [1, 100]$ over a 200×200 grid (5 m spatial resolution) as a superposition of 2D Gaussian radial basis functions:

$$H(x,y) = \text{clip}\Bigg(H_0 + \sum_{i\in\{\text{city,sub,ind}\}} A_i \exp\left[-\frac{(x-\mu_{x,i})^2}{2\sigma_{x,i}^2} - \frac{(y-\mu_{y,i})^2}{2\sigma_{y,i}^2}\right] - A_{\text{lake}}\exp\left[-\frac{\|\mathbf{x} - \boldsymbol{\mu}_{\text{lake}}\|_2^2}{2\sigma_{\text{lake}}^2}\right],\ 1,\ 100\Bigg), \tag{8}$$

where $H_0 = 15.0$ is the baseline rural harm floor, the primary metropolitan peak is centered strictly at the user's city coordinate $\boldsymbol{\mu}_{\text{city}} = \mathbf{x}_{\text{city}}$ with amplitude $A_{\text{city}} = 88.0$ and spread $\sigma_{\text{city}} = 175.0$ m (guaranteeing that $\mathbf{x}_{\text{city}}$ is the darkest crimson region on the map with $H(\mathbf{x}_{\text{city}}) = 100.0$), secondary suburban ($A_{\text{sub}} = 36.0$) and industrial ($A_{\text{ind}} = 25.0$) clusters lie on one flank of the flight corridor, and a negative Gaussian depression ($A_{\text{lake}} = 14.2$, $\sigma_{\text{lake}} = 140.0$ m) carves a low-harm open-water/field basin on the opposite flank. The defender selects $\mathbf{x}_{\text{safe}}$ by solving the constrained discrete minimization problem over the reachable diversion sector $\Omega_{\text{reachable}}$:

$$\mathbf{x}_{\text{safe}} = \arg \min_{(x,y)\in\Omega_{\text{reachable}}} H(x, y). \tag{9}$$

3 Mathematical Derivations and Theoretical Guarantees

To explain *why* the simulation behaves as it does during an academic defense, we present three self-contained mathematical derivations connecting the spoofing offset, the virtual ghost target, and the detector residual.

3.1 Derivation 1: Indirect Steering via the Virtual Ghost Target Transformation

A key question often asked in project reviews is: *"How can the defender steer the drone into the safe zone without hacking into its flight control software or motors?"*

Theorem 1 (Virtual Ghost Target Equivalence)**.** *Injecting an additive spoofing bias $+\boldsymbol{\delta}[k]$ into the GPS receiver of a line-of-sight guidance controller aiming at $\mathbf{x}_{\text{city}}$ is mathematically equivalent to leaving the GPS unspoofed while relocating the drone's physical target to a time-varying* ***Virtual Ghost Target****:*

$$\mathbf{x}_{\text{ghost}}[k] \triangleq \mathbf{x}_{\text{city}} - \boldsymbol{\delta}[k]. \tag{10}$$

Proof. Substitute the GPS measurement model (2) into the perceived line-of-sight aiming vector $\mathbf{e}_{\text{perc}}[k] = \mathbf{x}_{\text{city}} - \mathbf{x}_{\text{perceived}}[k]$ inside the guidance law (3):

$$\begin{aligned} \mathbf{e}_{\text{perc}}[k] &= \mathbf{x}_{\text{city}} - \big(\mathbf{x}_{\text{true}}[k] + \boldsymbol{\delta}[k] + \boldsymbol{\eta}[k]\big) \\ &= \underbrace{\big(\mathbf{x}_{\text{city}} - \boldsymbol{\delta}[k]\big)}_{\mathbf{x}_{\text{ghost}}[k]} - \mathbf{x}_{\text{true}}[k] - \boldsymbol{\eta}[k]. \end{aligned} \tag{11}$$

Taking the expectation $\mathbb{E}[\boldsymbol{\eta}[k]] = \mathbf{0}$, the drone's velocity vector (3) points strictly along the vector from its *true physical position* $\mathbf{x}_{\text{true}}[k]$ toward $\mathbf{x}_{\text{ghost}}[k] = \mathbf{x}_{\text{city}} - \boldsymbol{\delta}[k]$.

- At launch ($k = 0$), $\boldsymbol{\delta}[0] = \mathbf{0}$, so $\mathbf{x}_{\text{ghost}}[0] = \mathbf{x}_{\text{city}}$; the drone initially flies straight toward the city.
- As $\boldsymbol{\delta}[k]$ ramps toward $\boldsymbol{\delta}_{\text{final}} = \mathbf{x}_{\text{city}} - \mathbf{x}_{\text{safe}}$, the virtual ghost target slides smoothly across the map from $\mathbf{x}_{\text{city}}$ to:

$$\lim_{k\to\infty} \mathbf{x}_{\text{ghost}}[k] = \mathbf{x}_{\text{city}} - \boldsymbol{\delta}_{\text{final}} = \mathbf{x}_{\text{city}} - (\mathbf{x}_{\text{city}} - \mathbf{x}_{\text{safe}}) = \mathbf{x}_{\text{safe}}. \tag{12}$$

- Because the closed-loop guidance law is asymptotically stable ($s(d) > 0$ for all $d > 0$), $\mathbf{x}_{\text{true}}[k]$ tracks $\mathbf{x}_{\text{ghost}}[k]$ and converges to $\mathbf{x}_{\text{true}}(\infty) = \mathbf{x}_{\text{safe}}$, while the perceived position converges to $\mathbf{x}_{\text{perceived}}(\infty) = \mathbf{x}_{\text{safe}} + (\mathbf{x}_{\text{city}} - \mathbf{x}_{\text{safe}}) = \mathbf{x}_{\text{city}}$.

□

3.2 Derivation 2: Exact First-Order Recurrence for the Detector Residual

Next, we derive the exact discrete-time dynamics of the anomaly detector residual $r[k]$ to prove why a rate-limited ramp remains hidden below τ while a step jump triggers an immediate alarm.

Theorem 2 (Innovation Residual Dynamics and Steady-State Plateau)**.** *Let* $\mathbf{e}_{\text{res}}[k] \triangleq \mathbf{x}_{\text{perceived}}[k] - \mathbf{x}_{\text{imu}}[k]$ *denote the signed detector innovation vector, so that* $r[k] = \|\mathbf{e}_{\text{res}}[k]\|_2$*. Under the coupled dynamics* (1)*,* (2)*, and* (6)*,* $\mathbf{e}_{\text{res}}[k]$ *obeys the first-order linear recurrence:*

$$\mathbf{e}_{\text{res}}[k+1] = (1 - \alpha_{\text{INS}})\,\mathbf{e}_{\text{res}}[k] + \Delta\boldsymbol{\delta}[k] + \Delta\boldsymbol{\eta}[k] - \mathbf{w}_{\text{imu}}[k], \tag{13}$$

where $\Delta\boldsymbol{\delta}[k] = \boldsymbol{\delta}[k+1] - \boldsymbol{\delta}[k]$ *and* $\Delta\boldsymbol{\eta}[k] = \boldsymbol{\eta}[k+1] - \boldsymbol{\eta}[k]$*. Consequently, during a constant-rate spoofing ramp of step magnitude* $\|\Delta\boldsymbol{\delta}[k]\|_2 = \Delta\delta_{\max}$*, the expected residual converges monotonically to the finite steady-state plateau:*

$$r_{\text{ss}} = \frac{\Delta\delta_{\max}}{\alpha_{\text{INS}}}. \tag{14}$$

Proof. Write the innovation vector at step $k+1$ by substituting (2) and (1):

$$\begin{aligned}
\mathbf{x}_{\text{perceived}}[k+1] &= \mathbf{x}_{\text{true}}[k+1] + \boldsymbol{\delta}[k+1] + \boldsymbol{\eta}[k+1] \\
&= \mathbf{x}_{\text{true}}[k] + \mathbf{v}[k]\,\Delta t + \boldsymbol{\delta}[k] + \Delta\boldsymbol{\delta}[k] + \boldsymbol{\eta}[k] + \Delta\boldsymbol{\eta}[k] \\
&= \mathbf{x}_{\text{perceived}}[k] + \mathbf{v}[k]\,\Delta t + \Delta\boldsymbol{\delta}[k] + \Delta\boldsymbol{\eta}[k].
\end{aligned} \tag{15}$$

Subtract the IMU update equation (6) from (15):

$$\begin{aligned}
\mathbf{e}_{\text{res}}[k+1] &= \mathbf{x}_{\text{perceived}}[k+1] - \mathbf{x}_{\text{imu}}[k+1] \\
&= (\mathbf{x}_{\text{perceived}}[k] - \mathbf{x}_{\text{imu}}[k]) + (\mathbf{v}[k]\,\Delta t - \mathbf{v}[k]\,\Delta t) \\
&\quad - \alpha_{\text{INS}}(\mathbf{x}_{\text{perceived}}[k] - \mathbf{x}_{\text{imu}}[k]) + \Delta\boldsymbol{\delta}[k] + \Delta\boldsymbol{\eta}[k] - \mathbf{w}_{\text{imu}}[k] \\
&= (1 - \alpha_{\text{INS}})\,\mathbf{e}_{\text{res}}[k] + \Delta\boldsymbol{\delta}[k] + \Delta\boldsymbol{\eta}[k] - \mathbf{w}_{\text{imu}}[k].
\end{aligned} \tag{16}$$

Notice that the commanded velocity term $\mathbf{v}[k]\,\Delta t$ *cancels out identically*! Taking the deterministic part (dropping zero-mean noise) with initial condition $\mathbf{e}_{\text{res}}[0] = \mathbf{0}$ and constant step vector $\Delta\boldsymbol{\delta}[k] = \Delta\delta_{\max}\,\mathbf{u}_\delta$ for $0 \le k < K_{\text{ramp}}$, unrolling the geometric series yields the closed-form time response:

$$\|\mathbf{e}_{\text{res}}[k]\|_2 = \Delta\delta_{\max} \sum_{j=0}^{k-1} (1 - \alpha_{\text{INS}})^j = \frac{\Delta\delta_{\max}}{\alpha_{\text{INS}}}\Big(1 - (1 - \alpha_{\text{INS}})^k\Big). \tag{17}$$

As k increases during the ramp, $(1 - \alpha_{\text{INS}})^k \to 0$, proving $\lim_{k\to\infty} \|\mathbf{e}_{\text{res}}[k]\|_2 = \Delta\delta_{\max}/\alpha_{\text{INS}}$. Once the ramp saturates at $\boldsymbol{\delta}_{\text{final}}$ ($k \ge K_{\text{ramp}}$), $\Delta\boldsymbol{\delta}[k] = \mathbf{0}$ and the residual decays exponentially to zero as $r[k] \propto (1 - \alpha_{\text{INS}})^{k - K_{\text{ramp}}}$. □

3.3 Derivation 3: Analytical Bound for Rate-Limit Optimization (Equation 7)

From Theorem 2, to guarantee that the peak residual never exceeds a safety factor $\gamma \in (0, 1)$ times the threshold τ (we use $\gamma = 0.95$, i.e., a 5% margin for stochastic noise $\Delta\boldsymbol{\eta}[k] - \mathbf{w}_{\text{imu}}[k]$), the stealth ramp rate must satisfy:

$$r_{\text{ss}} = \frac{\Delta\delta_{\max}}{\alpha_{\text{INS}}} \le \gamma\,\tau \implies \Delta\delta_{\max} \le \gamma\,\alpha_{\text{INS}}\,\tau. \tag{18}$$

Substituting our simulation parameters ($\alpha_{\text{INS}} = 0.068$, $\tau = 90.0$ m, $\gamma \approx 0.91$ after accounting for the ~ 2.2 m stochastic noise crest) yields:

$$\Delta\delta^*_{\max} \approx 0.068 \times 81.9 \text{ m} = 5.57 \text{ m/step} \quad \left(v^*_\delta = \frac{5.57 \text{ m}}{0.5 \text{ s}} = 11.14 \text{ m/s}\right), \tag{19}$$

which matches the numerical rate-sweep optimizer (`optimize_delta_rate_max`) to two decimal places! Conversely, in **Case C (Aggressive Step Jump)**, the entire offset $\|\boldsymbol{\delta}_{\text{final}}\|_2 = 488.48$ m is applied in a single step ($\|\Delta\boldsymbol{\delta}[k_{\text{jump}}]\|_2 = 488.48$ m), causing (13) to jump instantaneously to $r[k_{\text{jump}} + 1] \approx 490.53 \text{ m} \gg \tau = 90.0$ m.

4 Simulation Experiments and Quantitative Results (What We Achieved)

To validate the mathematical framework, we executed four comprehensive simulation scenarios in `simulation.py` using fixed random seed `RANDOM_SEED = 42` for exact reproducibility. Table 1 summarizes the physical and sensor parameters used across all runs.

Table 1: Global Simulation Parameters and Sensor Specifications

Parameter Symbol	Configured Value	Physical Meaning in Simulation
$\mathbf{x}_{\text{start},1}$, $\mathbf{x}_{\text{start},2}$	$[100, 100]$ m, $[60, 240]$ m	User-configurable launch coordinates for Enemy Drones 1 & 2
$\mathbf{x}_{\text{city}}$	[800.0, 800.0] m	User-configurable Protected City center (Peak Harm 100.0/100)
$\mathbf{x}_{\text{safe}}$	[794.0, 311.6] m	Auto-selected minimum-damage safe zone (Harm 3.42/100)
$\boldsymbol{\delta}_{\text{final}}$	[+6.0, +488.4] m	Required terminal spoofing offset ($\|\boldsymbol{\delta}_{\text{final}}\|_2 = 488.48$ m)
$v_{\max}$, Δt	20.0 m/s, 0.5 s	Maximum drone cruise speed and discrete simulation time step
σ_{gps}, σ_{imu}	1.2 m, 0.25 m	GPS receiver noise std dev and IMU dead-reckoning drift std dev
α_{INS}, τ	0.068, 90.0 m	Aided INS complementary filter gain and anomaly threshold
$\Delta\delta^*_{\max}$	5.57 m/step (11.14 m/s)	Optimal stealth spoofing ramp rate from Eq. 7 sweep

4.1 Comparative Performance Across All Four Simulation Cases

Table 2 reports the quantitative performance metrics recorded by `run_simulation()` for the default coordinate configuration ($\mathbf{x}_{\text{start}} = [100, 100]$ m, $\mathbf{x}_{\text{city}} = [800, 800]$ m), alongside a second verification run using custom user-entered coordinates ($\mathbf{x}_{\text{start}} = [120, 180]$ m, $\mathbf{x}_{\text{city}} = [780, 820]$ m).

Table 2: Quantitative Summary of Simulation Cases A, B, C, Multi-Drone Extension, and Custom User Coordinates

Simulation Scenario	Start $\mathbf{x}_0$ [m]	Final True Pos [m]	Miss Dist (Safe)	Divert Time	Max $r[k]$ [m]	Detector Status ($r[k] \lessgtr \tau$)
Case A: No Spoofing (Baseline)	[100.0, 100.0]	[800.3, 799.7]	488.20 m	49.0 s	4.68 m	NO ALARM (Hits City)
Case B: Stealth Spoof (Drone 1)	[100.0, 100.0]	[794.3, 311.3]	**0.40** m	44.5 s	**84.12** m	NO ALARM (Stealth OK)
Case C: Aggressive Step Jump	[100.0, 100.0]	[794.3, 311.3]	0.40 m	38.5 s	**490.53** m	DETECTED at $t = 12.5$ s
Extension: Stealth Spoof (Drone 2)	[60.0, 240.0]	[758.9, 337.0]	**0.43** m	42.5 s	**83.20** m	NO ALARM (Stealth OK)
Custom User Input Run (Case B)	[120.0, 180.0]	[789.2, 341.4]	**0.40** m	43.5 s	**84.10** m	NO ALARM (Stealth OK)

4.2 Detailed Interpretation of the Five Generated Visual Figures

When presenting the simulation outputs saved in `./outputs/` to the evaluation committee, each figure illustrates a specific theoretical mechanism:

1. **Animated 2D Map (`fig1_drone_spoofing_animation.gif`)**: Visualizes the real-time divergence between the drone's *perceived* trajectory (dashed orange line heading straight into the red star at $\mathbf{x}_{\text{city}}$) and its *true* physical trajectory (solid blue curve bending southward into the green minimum-harm basin $\mathbf{x}_{\text{safe}}$). The live telemetry badge confirms that $r[k]$ stays below $\tau = 90$ m throughout the entire diversion.

2. **Static Trajectory Comparison & 2-Drone Extension (`fig2_trajectories_comparison.png`)**: The left panel contrasts Case A (straight 45° path into the city), Case B (smooth stealth diversion to $[794.0, 311.6]$ m), and Case C (abrupt turn at $t = 12.0$ s accompanied by an immediate detection alarm marker at $t = 12.5$ s). The right panel proves that the defense scales to a 2-drone attack formation by assigning separated landing slots within the same low-harm basin.

3. **Spoofing Offset Magnitude $\|\boldsymbol{\delta}[k]\|_2$ vs. Time (`fig3_spoofing_offset_norm.png`)**: Validates Equation (5). Case B exhibits a bounded linear ramp at slope 11.14 m/s from 0 m to $\|\boldsymbol{\delta}_{\text{final}}\|_2 = 488.48$ m at $t \approx 44$ s, whereas Case C exhibits a discontinuous vertical step at $t = 12$ s.

4. **Detector Residual $r[k]$ & Rate Optimization Sweep (`fig4_detector_residuals.png`)**: Validates Theorem 2 ((17)–(14)). In the left panel, Case B rises along the theoretical exponential curve $(1-0.068)^k$ to a plateau of 84.12 m $< \tau = 90$ m before decaying exponentially back to zero after $t = 44$ s, whereas Case C pierces the red alarm region at 490.53 m. The right panel plots peak residual $\max_k r[k]$ versus ramp rate v_δ, showing the exact linear slope $1/\alpha_{\text{INS}}$ predicted by (18).

5. **True Separation Distances Over Time (`fig5_distances_to_targets.png`)**: Plots $\|\mathbf{x}_{\text{true}}[k] - \mathbf{x}_{\text{city}}\|_2$ and $\|\mathbf{x}_{\text{true}}[k] - \mathbf{x}_{\text{safe}}\|_2$. Under stealth spoofing, the drone approaches no closer than ~ 355 m to the city before peeling away and stabilizing at 488.2 m from the city and 0.40 m from the safe zone.

5 Conclusion and Future Scope

This project demonstrated mathematically and experimentally that defensive GPS spoofing can covertly redirect autonomous GPS-guided drones away from high-density population centers into minimum-harm safe zones without triggering onboard IMU/GPS residual detectors. By proving the Virtual Ghost Target equivalence ($\mathbf{x}_{\text{ghost}}[k] = \mathbf{x}_{\text{city}} - \boldsymbol{\delta}[k]$) and deriving the exact steady-state residual law ($r_{\text{ss}} = \Delta\delta_{\max}/\alpha_{\text{INS}}$), we transformed spoofing rate selection from heuristic trial-and-error into a closed-form constrained optimization problem. Future extensions of this work include extending the 2D point-mass kinematics to 6-DOF rigid-body dynamics with wind gusts and analyzing tightly-coupled Extended Kalman Filter (EKF) detectors with visual-odometry aiding.

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